CardDemo: COBOL → Rust

Project progress report: session 1

Prepared with Claude Code (Opus 5.5)
2026-11-05

Status at a glance

Where we are

Done in session 1

- Requirements specification (PDF, v1.1)
- Rust workspace, 6 crates, **all 30 base programs**
  - 17 CICS online programs + terminal UI
  - 13 batch programs + JCL job runner
- Manual test suite (PDF, 40 cases, 100% FR-group coverage)
- Live-demo script, CLI, one-command reset

Open

- Differential tests against GnuCOBOL: *blocked*, the `gmp-devel` headers are missing
- Optional modules (IMS/DB2/MQ): out of scope by decision
- Presentation of session 2 results

Scope and approach

Scope (agreed 2026-11-05)

In scope	Out of scope
Base app: COBOL, CICS, VSAM, JCL, assembler	Authorisations (IMS / DB2 / MQ)
20.6k lines COBOL, 2.8k copybook, 4.5k BMS	Transaction types (DB2)
Terminal UI (3270 style), SQLite, fixed-width I/O	MQ account and date inquiry

Decisions: faithful translation (same messages, layouts, bytes, **and defects**); GnuCOBOL for batch differential tests; live demo planned.

Architecture

Crate	Replaces	L
`cobol`	COBOL data semantics: zoned, COMP-3, edit pictures, cp037 + generated copybooks	1
`vsam`	VSAM KSDS/AIX on SQLite: EBCDIC key order, CICS browse rules	
`cics`	BMS maps parsed from the original sources; XCTL / RETURN, file control, TDQ	
`online`	the 17 CICS programs, one function per COBOL paragraph	4
`batch`	13 batch programs + JCL jobs (SORT, REPRO, GDG, internal reader)	1
`carddemo`	binary: ratatui TUI, `job`, `batch`, `reset`, `script`	

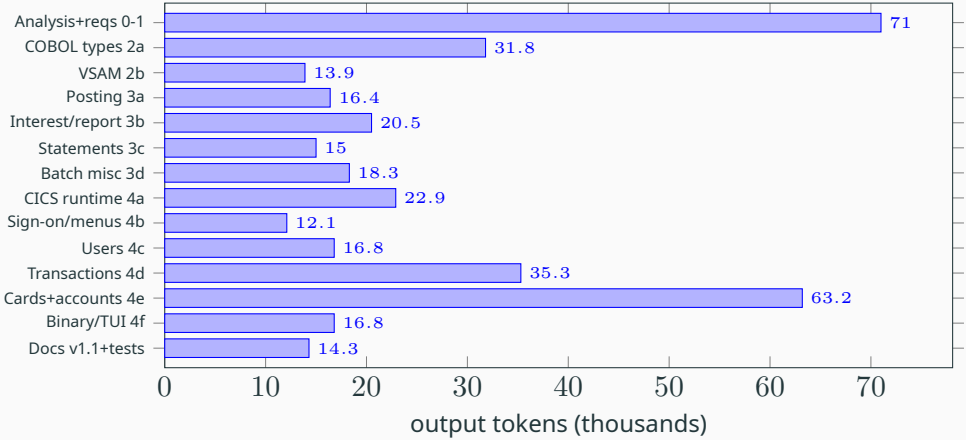
About **9.7k lines of Rust** (plus 1.1k generated) replace about **27k lines** of COBOL, copybooks and BMS.

Plan versus actual

Phases

#	Phase	Plan	Actual
0	Inventory	call graph, CRUD matrix	done, scripted extraction
1	Requirements PDF	FR / NFR / traceability	26 pp, 165 FR, 12 NFR, 10 KD
2	Foundation crates	types, records, store	done; records generated from copybooks
3	Batch	13 programs + jobs	done; export/import round trip byte-identical
4	Online + TUI	17 programs	done; scripted terminal journeys pass
5	Optional modules	—	dropped (scope decision)
6	Manual test PDF	per requirement	40 cases, 122 steps
7	Hardening, demo	runbook	demo script, README; diff tests open
8	Progress deck	this document	—

Effort by phase (output tokens)



Time and tokens (from the Claude Code transcripts)

Metric	Value
Active session time (gaps > 10 min excluded)	≈ 1.1 h
Assistant turns / tool calls	194 / 212
Output tokens	367k
Input tokens (uncached)	0.4k
Prompt cache writes / reads	0.76M / 90.9M
Peak context window fill	≈ 80% of 1M
Commits on branch `rust-port`	15

Measured with `carddemo-rs/tools/usage.py` and `tools/phase_metrics.py`. Cache reads dominate: every turn re-reads the conversation.

Quality and fidelity

Testing

- **Unit tests:** numerics (zoned/COMP-3/binary), edit pictures, NUMVAL, cp037, VSAM browse rules, BMS parser, CEEDAYS dates
- **Record tests:** generated layouts decode the original EBCDIC sample data
- **Batch:** the full nightly cycle (TRANBKP → POSTTRAN → INTCALC → COMBTRAN → TRANREPT → CREASTMT); CBEXPORT → CBIMPORT is byte-identical
- **Online:** scripted 3270 journeys through all 17 programs (sign-on, admin, lists, updates, payments, report submission)
- **Result:** all tests pass (`cargo test --workspace`)
- **Pending:** byte-level comparison with the original batch programs under GnuCOBOL

Defects found in the original (kept on purpose)

ID	Where	Effect
KD-01	INTCALC	the last account's interest is never written back
KD-02/03	TRANREPT	last amount counted twice; no final account total
KD-07	card update	the CVV is reset to 000
KD-08	account update	the "blank phone" test checks the area code twice
KD-10	user/tran list	search past the last key $\rightarrow$ AEIP abend
KD-11	user/tran list	PF7 then PF8 on page 1 $\rightarrow$ empty page

Also: the COACTVW map blanks most of the `Tran:` header, and the sample data fails the COACTUPC state/zip check for 48 of 50 customers.

Lessons learned

- **Read the source, never a filtered view.** A space-collapsing reader and a grep filter both produced wrong requirements (caught and fixed: KD-06 was withdrawn).
- **COBOL semantics hide in ordering.** For example `SET NEXT-PAGE-NO` *before* the commarea MOVE.
- **The 3270 matters:** FSET fields, nulls versus spaces, overlapping map fields.
- **Generate from the originals:** copybooks → Rust records, BMS → screens, CSLKPCDY → lookups.

Next steps

Session 2 plan

1. Install `gmp-devel` → GnuCOBOL differential tests for POSTTRAN, INTCALC, TRANREPT, CREASTMT (and verify the NEXT SENTENCE semantics in CBTRN03C)
2. Execute the manual test suite; record results in the execution log
3. Hardening: clippy, multi-terminal locking test, EBCDIC export of the export file
4. Live demo rehearsal (`docs/DEMO.md`)
5. Updated deck with measured results of session 2

Demo

```
carddemo reset          # 5 ms
carddemo online         # sign on as USER0001 / PASSWORD
carddemo batch          # nightly cycle in about 20 ms
```

Branch `rust-port`: `docs/requirements.pdf`, `docs/manual-test-suite.pdf`, `docs/DEMO.md`.